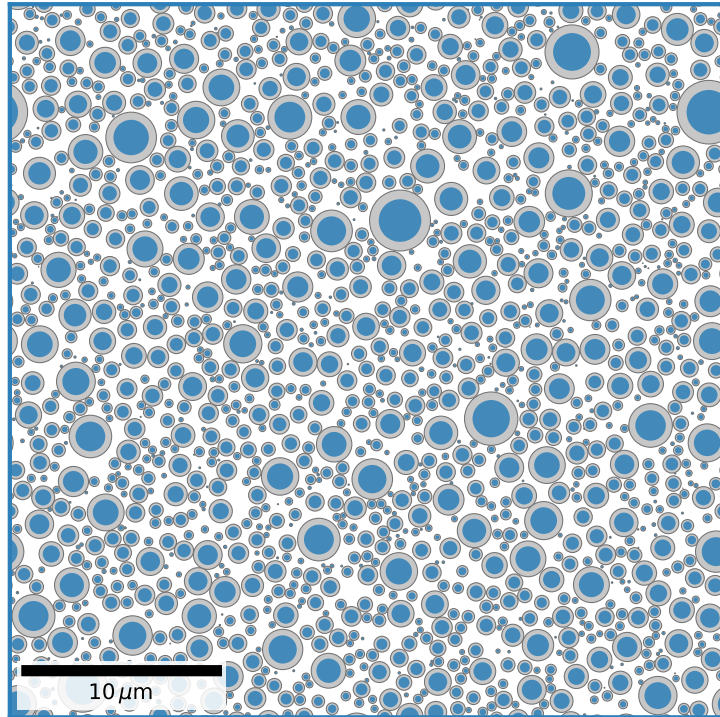


SMALL-CALIBRE (CC genu / general WM)

$\langle d \rangle = 0.61 \mu\text{m}$, $\langle S/V \rangle = 6.2 \mu\text{m}^{-1}$

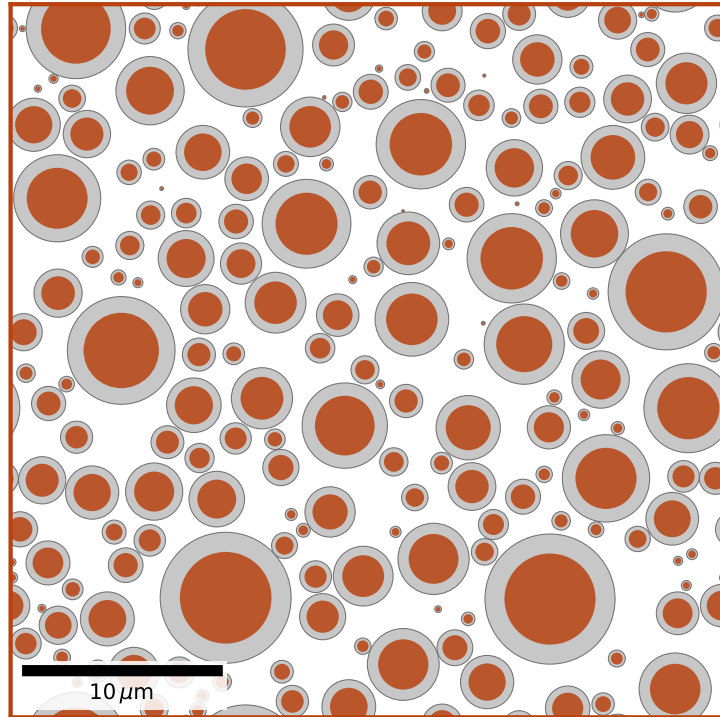
recovered MWF = 14.0% (+0.6 pp)



LARGE-CALIBRE (corticospinal / CC midbody)

$\langle d \rangle = 2.00 \mu\text{m}$, $\langle S/V \rangle = 1.9 \mu\text{m}^{-1}$

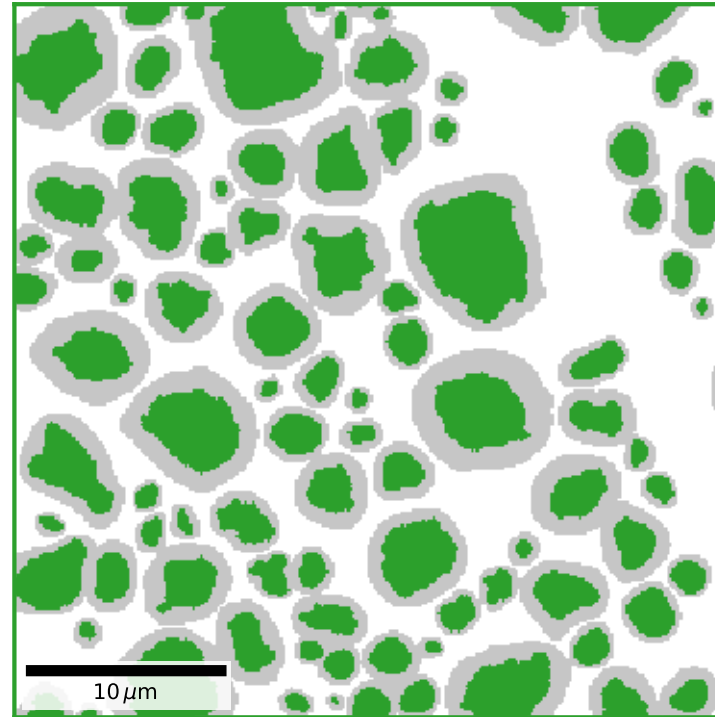
recovered MWF = 13.5% (-0.0 pp)



REAL segmentation (rat spinal cord (AxonDeepSeg))

$\langle d \rangle = 2.00 \mu\text{m}$, $S/V = 1.9 \mu\text{m}^{-1}$

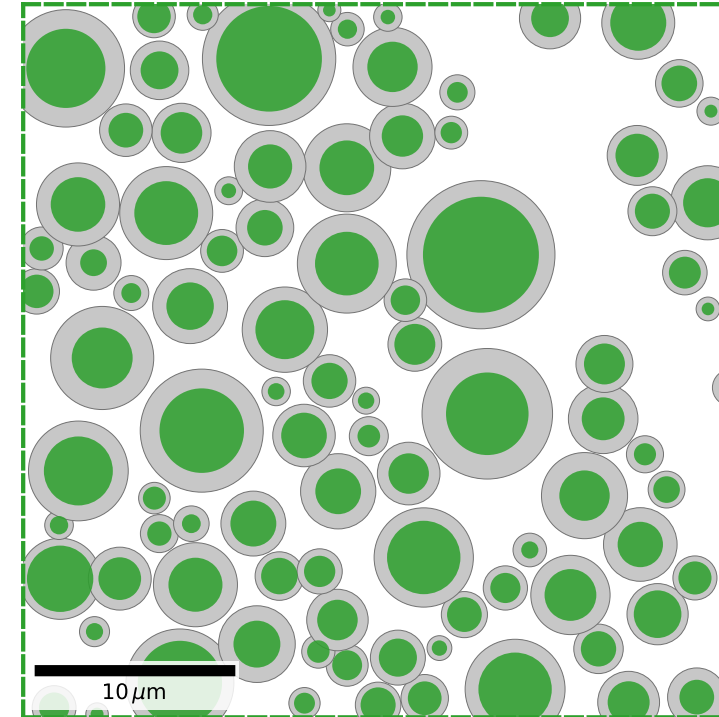
real EM (MVF = 0.32)



REAL fitted cylinders

$\langle d \rangle = 2.00 \mu\text{m}$, **per-axon $g = 0.66$**

(what the walk sees)



Panels 1--2: synthetic packs at identical myelin volume ($36 \mu\text{m}$ cell, $f = 0.55$, $g = 0.70$) $\Rightarrow$ higher S/V (smaller axons) gives larger $\rho(S/V)$ in T_2 and lower recovered MWF at constant myelin. Panels 3--4: a real rat cross-section and the myelinated cylinders fitted to it (real positions, per-axon diameter and g -ratio) that the Monte Carlo walks.